

Technology Stack

Date	15 March 2026
Team ID	SWTID-2026-9392
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/ CSS)	React.js / Next.js, Tailwind CSS	Provides a responsive, dynamic user interface for interactive comic panel layout, real-time previewing, and component reusability.
2	Backend / Server-Side (e.g., Node.js, Python/ Django, Java)	Python (FastAPI / Flask)	Offers high-performance asynchronous request handling and native integration with AI/ML SDKs, enabling fast prompt processing and backend API routing.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MongoDB / PostgreSQL	Handles unstructured comic metadata (panel layouts, story script schemas, user sessions) and JSON document payloads efficiently.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Docker, Vercel / AWS EC2	Ensures containerized deployment, quick horizontal scaling under peak traffic, and cost-effective cloud hosting.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub & GitHub Actions	Enables seamless team collaboration, automated build pipelines, continuous integration, and streamlined testing before release.
6	Third-Party APIs / Other Tools		Provides state-of-the-art Generative AI for generating comic story scripts and prompts,

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	(e.g., Stripe, Twilio, Google Maps)	Google Gemini API (Gemini Flash/Pro) & ReportLab / PDFkit	combined with PDF compilation libraries for seamless file exporting.